

Response to Reviewer 1

Manuscript **TP-1083** — *The Incommensurability Principle in Biological Transport
Transport Phenomena* (De Gruyter Brill)

We thank Reviewer 1 for the careful reading and for an observation that improved the framing of the paper. Below we reproduce the comment (paraphrased from the report) and detail our response. The highlighted copy uses a distinct colour per source, as the editor requested: blue for Reviewer 1, red for Reviewer 2, green for editorial requests, and magenta for author-initiated refinements.

Comment 1 — Opening contrast (mouse vs. whale heart rate).

Reviewer comment.

The manuscript is well written and the concepts are well explained. The topic is interesting and will attract Transport Phenomena readers. The introduction contrasts the mouse (~ 600 bpm) and the whale (~ 20 bpm) heart rates. Could the comparison be made in terms of the volume of blood pumped rather than the heart rate?

Response. We agree. We have replaced the heart-rate contrast with a comparison based on stroke volume and cardiac output. Cardiac output scales as $\dot{Q} \propto M^{3/4}$ and stroke volume as $\propto M$, so a single whale contraction ejects a blood volume roughly seven orders of magnitude larger than the mouse's. That volume travels through vessels whose radius $R \propto M^{3/8}$ sets the Womersley number $Wo \propto M^{1/4}$ (derived in Section 6)—making the whale's vasculature the more inertial of the two despite its slower beat, which is precisely the point. The cardiac-output allometry is now cited to Noujaim et al. (*Circulation* **110**, 2004, “From Mouse to Whale”).
Location: Section 1, blue.

Additional author-initiated refinements. In the same highlighted copy, we also mark in magenta a small set of author-initiated refinements made during final checking: the characteristic-admittance notation is written as $Y_c \propto \sqrt{i\omega\rho Y_L}$, equivalently $Y_c \propto \sqrt{1 - F_{10}}$; the modified Womersley admittance convention is made explicit as $M'_{10} = -(8i/Wo^2)(1 - F_{10})$; the minimax coordinate is distinguished from the cardiac duty cycle and set on $\eta \in [0, 1]$; and several overstrong phrases were softened without changing the quantitative predictions.